

CS 228 : Logic in Computer Science

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Satisfiability, Validity and Equivalence

- ▶ A formula φ over a signature τ is said to be **satisfiable** iff for some τ -structure $\mathcal{A}$ and assignment α , $\mathcal{A} \models_{\alpha} \varphi$
- ▶ A formula φ over a signature τ is said to be **valid** iff for every τ -structure $\mathcal{A}$ and assignment α , $\mathcal{A} \models_{\alpha} \varphi$
- ▶ Formulae $\varphi(x_1, \dots, x_n)$ and $\psi(x_1, \dots, x_n)$ are **equivalent** denoted $\varphi \equiv \psi$ iff for every $\mathcal{A}$ and α , $\mathcal{A} \models_{\alpha} \varphi$ iff $\mathcal{A} \models_{\alpha} \psi$

Equisatisfiability

Let $\varphi_1(x) = \forall y R(x, y)$ and $\varphi_2 = \exists x \forall y R(x, y)$.

- ▶ It is clear that whenever $\mathcal{A} \models \varphi_2$, one can find an assignment α such that $\mathcal{A} \models_{\alpha} \varphi_1(x)$.
- ▶ Likewise, if $\mathcal{A} \models_{\alpha} \varphi_1(x)$, then $\mathcal{A} \models \varphi_2$.
- ▶ Thus, $\varphi_1(x), \varphi_2$ are **equisatisfiable**.

True or False?

For a formula φ and assignments α_1 and α_2 such that for every $x \in \text{free}(\varphi)$, $\alpha_1(x) = \alpha_2(x)$, $\mathcal{A} \models_{\alpha_1} \varphi$ iff $\mathcal{A} \models_{\alpha_2} \varphi$

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- For example, $\varphi(y) = \forall x(R(x, y) \rightarrow \forall zP(z))$

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- ▶ Consider two assignments α_1, α_2 such that $\alpha_1(y) = \alpha_2(y) = \alpha(\text{say})$

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- ▶ Evaluate for all $a, b \in u(\mathcal{A})$, $R(a, \alpha) \rightarrow P(b)$
- ▶ $\mathcal{A} \models_{\alpha_1} \varphi$ iff $\mathcal{A} \models_{\alpha_2} \varphi$

True or False?

For a sentence φ , and any two assignments α_1 and α_2 , $\mathcal{A} \models_{\alpha_1} \varphi$ iff $\mathcal{A} \models_{\alpha_2} \varphi$

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No free variables!

Check Satisfiability

Let τ be a signature with a single unary relation P . Consider the structure $\mathcal{A} = (U_{\mathcal{A}} = \{0, 1\}, P^{\mathcal{A}} = \{1\})$.

Let $\varphi = \forall x_1 \forall x_2 \dots \forall x_n (P(x_1) \rightarrow (P(x_2) \rightarrow (P(x_3) \dots \rightarrow (P(x_n) \rightarrow P(x_1)) \dots)))$.

Does $\mathcal{A} \models \varphi$?

Check Satisfiability

Let $\varphi(y) = \exists x(E(x, y) \wedge \neg(y = x) \wedge \forall z[E(z, y) \rightarrow z = x])$ over the signature τ containing a binary relation E . Is $\varphi(y)$ satisfiable under some graph structure?

Check Satisfiability

Let $\psi(z) = \exists x [Q_a(x) \wedge \forall y [(y \leq x \wedge Q_b(y)) \rightarrow (z < x \wedge y < z \wedge Q_c(z))]]$
over the signature τ having the relational symbols $<$, Q_a , Q_b , Q_c and
unary function S . Does $\psi(z)$ evaluate to true under some word
structure?

Check Satisfiability

Let $\zeta = P(0) \wedge \forall x(P(x) \rightarrow P(S(x))) \wedge \exists x \neg P(x)$ over a signature τ containing the constant 0, unary function S and unary relation P .
Is ζ satisfiable?

Normal Forms in FOL

Recap : Satisfaction

We define the relation $\mathcal{A} \models_{\alpha} \varphi$ (read as φ is true in $\mathcal{A}$ under the assignment α) inductively:

- ▶ $\mathcal{A} \not\models_{\alpha} \perp$
- ▶ $\mathcal{A} \models_{\alpha} t_1 = t_2$ iff $\alpha(t_1) = \alpha(t_2)$
- ▶ $\mathcal{A} \models_{\alpha} R(t_1, \dots, t_k)$ iff $(\alpha(t_1), \dots, \alpha(t_k)) \in R^{\mathcal{A}}$
- ▶ $\mathcal{A} \models_{\alpha} (\varphi \rightarrow \psi)$ iff $\mathcal{A} \not\models_{\alpha} \varphi$ or $\mathcal{A} \models_{\alpha} \psi$
- ▶ $\mathcal{A} \models_{\alpha} (\forall x)\varphi$ iff for every $a \in u(\mathcal{A})$, $\mathcal{A} \models_{\alpha[x \mapsto a]} \varphi$
- ▶ $\mathcal{A} \models_{\alpha} (\exists x)\varphi$ iff there is some $a \in u(\mathcal{A})$, $\mathcal{A} \models_{\alpha[x \mapsto a]} \varphi$

Last two cases, α has no effect on the value of x . Thus, assignments matter **only** to free variables.

Equivalences

Let F, G be arbitrary FOL formulae.

$$1. \neg \forall x F \equiv \exists x \neg F$$

$$2. \neg \exists x F \equiv \forall x \neg F$$

$\mathcal{A} \models_{\alpha} \neg \forall x F$ iff

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$$\begin{aligned} \mathcal{A} \models_{\alpha} \neg \forall x F &\text{ iff } \mathcal{A} \not\models_{\alpha} \forall x F \\ &\text{ iff } \mathcal{A} \not\models_{\alpha[x \mapsto a]} F \text{ for some } a \in U^{\mathcal{A}} \end{aligned}$$

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Equivalences

If x does not occur free in G then

1. $(\forall x F \wedge G) \equiv \forall x (F \wedge G)$
2. $(\forall x F \vee G) \equiv \forall x (F \vee G)$
3. $(\exists x F \wedge G) \equiv \exists x (F \wedge G)$
4. $(\exists x F \vee G) \equiv \exists x (F \vee G)$

$\mathcal{A} \models_{\alpha} \forall x F \wedge G$ iff $\mathcal{A} \models_{\alpha} \forall x F$ and $\mathcal{A} \models_{\alpha} G$

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$\mathcal{A} \models_{\alpha} \forall x F \wedge G$ iff $\mathcal{A} \models_{\alpha} \forall x F$ and $\mathcal{A} \models_{\alpha} G$
iff for all $a \in U^{\mathcal{A}}$, $\mathcal{A} \models_{\alpha[x \mapsto a]} F$ and $\mathcal{A} \models_{\alpha} G$

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iff $\mathcal{A} \models \forall x (F \wedge G)$

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Let F, G be arbitrary FOL formulae.

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